MTH220 2023 S1 TOA

Feel free to contact me at maths_and_programming@outlook.sg if you need tuition. I provide university level tuition for the following SUSS units:

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* ICT162 Object Oriented Programming
* MTD215 Application of C++ in Multimedia
* ICT231 Data Structures and Algorithms I
* ICT232 Data Structures and Algorithms II
* MTH105 Fundamentals of Mathematics
* MTH107 Calculus I
* MTH108 Calculus II
* MTH316 Multivariable Calculus
* MTH207 Linear Algebra
* MTH219 Fundamentals of Statistics and Probability
* MTH220 Statistical Methods and Inference

Donaldson Tan

1st 1 00
2nd 1 30
3rd 2 00
4th 2 30

1st 1pm
2nd 1 30
3rd 2pm
4th 2 30

SUSS
SINGAPORE UNIVERSITY
OF SOCIAL SCIENCES

MTH220

Examination - July Semester 2023

Statistical Methods and Inference

Thursday, 23 November 2023 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FOUR (4)** questions and comprises **SIX (6)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.
4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1

You have been provided with a dataset containing information on the mileage of cars (in kilometers) along with their prices (in SGD). Your task is to build a simple linear regression model to predict the price of a car based on its mileage.

Car	X: Mileage (in km)	Y: Price ($)
1	10000	175000
2	30000	140000
3	50000	126000
4	15000	161000
5	40000	133000
6	A	119000
7	70000	112000
8	80000	B
9	90000	98000
10	100000	91000

Question 1a

We know that the mean mileage of the ten cars is 54500 km, and the mean price of the ten cars is $126000. Calculate the missing values of A and B.

(4 marks)

Question 1b

Determine the coefficients of the simple regression model and write down the full equation. Interpret the estimates.

(7 marks)

Question 1c

Predict the price for a car with 75000 km of mileage.

(3 marks)

Question 1d

Calculate the Pearson Correlation Coefficient between x and y. Comment on your findings.

(6 marks)

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Question 2

A researcher wants to compare the performance of students in online and traditional classrooms. The researcher collects two independent samples of students, one from an online classroom and the other from a traditional classroom. The scores of a math quiz for the students are recorded.

Online Classroom Scores: 48, 56, 58, 62, 61, 54, 58, 64, 65, 64, 63, 65

Traditional Classroom Scores: 66, 66, 65, 62, 62, 63, 68, 63, 69, 56, 75, 58

Question 2a

State the null and alternative hypotheses for testing whether there is a significant difference in the average math quiz scores between students in online and traditional classrooms.

(2 marks)

Question 2b

Apply a two-sample t-test assuming equal variance to determine if there is enough evidence to support the claim that there is a significant difference in average math quiz scores between the two classrooms. Use a significance level of 0.05.

(10 marks)

Question 2c

Comment briefly on the design of the study.

(3 marks)

Question 2d

Imagine that upon further investigation and with a larger sample size, it is revealed that there was, in fact, no difference in the scores of the math quiz between the two classrooms. Describe the type of error that has occurred in this case, discuss the potential impact of making such an error in educational decision-making and how can it be minimized.

(5 marks)

Question 3

The Ministry of Health is investigating the frequency of unplanned hospital admissions at a single hospital in a particular region. The hospital has recorded the number of unplanned admissions each day for a period of 365 days. The Ministry suspects that the number of unplanned hospital admissions follows a Poisson distribution.

You are provided with the following frequency table for the number of unplanned hospital admissions over the 365-day period:

Number of Unplanned Admissions (x)	Observed Frequency, in days (f)
0	8
1	17
2	48
3	60
4	71
5	61
6	39
7	26
8	20
9	10
10	3
11	1
12	1

Question 3a

State the null and alternative hypotheses for the Chi-square goodness-of-fit test to investigate whether the number of unplanned hospital admissions follows a Poisson distribution.

(2 marks)

Question 3b

Apply a Chi-square goodness-of-fit test to determine if there is enough evidence to support the claim that the number of unplanned hospital admissions fits the Poisson distribution by computing the expected frequencies. Use a significance level of 0.05.

(13 marks)

Question 3c

Comment on the result of the Chi-square test in the context of the problem. Based on the test result, what conclusions can be drawn about the distribution of unplanned hospital admissions at this hospital?

(3 marks)

Question 3d

Describe the assumptions and conditions required for the Chi-square goodness-of-fit test to be valid in this scenario. How can the Ministry ensure that these conditions are met during the investigation?

(6 marks)

Question 3e

With the help of the central limit theorem, compute the probability $P(\bar{X} > 4.65)$, where X is the number of unplanned hospital admissions per day.

(6 marks)

Question 4a

We have a deck of poker cards, from A, 2, 3,…, J, Q, K. Ignoring joker cards, we have 52 cards. The value of each card goes from Ace = 1, Twos = 2, …, Jacks = 11, Queens = 12, Kings = 13.

Let X be the random variable representing the value of randomly drawing a single card. After drawing a card, the card will be returned to the deck. (i.e. it is drawn with replacement).

Question 4a(i)

Calculate the expected value and the variance of X.

(3 marks)

Question 4a(ii)

Let's say there is another game where n cards are drawn with replacement (with replacement means that after a card is selected, it is returned to the deck before the next selection). Let T_n be the sum of the value of n card draws. Calculate the mean and variance and describe the distribution of T_n.

(3 marks)

Question 4a(iii)

What is the minimum number of n such that the probability of $\bar{X} = \frac{T_n}{n} > 7.5$ is less than 5%?

$P(\bar{X}$

(4 marks)

Question 4b

Suppose a customer support center receives 50 phone calls during a specific hour. The probability of a customer's call being resolved within 5 minutes is 0.6 (60%). The number of calls resolved within 5 minutes can be modeled by a binomial distribution.

Question 4b(i)

Using the exact binomial model, compute the probability of resolving exactly 30 calls within 5 minutes.

(2 marks)

Question 4b(ii)

Apply the normal approximation method, compute and estimate the probability of resolving exactly 30 calls within 5 minutes. Verify the conditions for which the normal approximation to the binomial will be appropriate and comment on your answer.

(5 marks)

Question 4b(iii)

Apply the normal approximation method, compute and estimate the probability that the support center resolves fewer than 25 calls within 5 minutes.

(3 marks)

Question 4c $Sd =$

A company claims that their new energy drink increases productivity. You conduct a study and find that the average productivity score of a random sample of 50 employees who consumed the drink is 85, with a standard deviation of 6. The mean productivity score without the drink is 80. Conduct a hypothesis test to determine if there is sufficient evidence to support the company's claim at a significance level of 0.05.

Sd

(10 marks)

----- END OF PAPER -----

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1(a) $\bar{x} = 54500 \quad \bar{y} = 126000 \quad n = 10$

$\Sigma x = n\bar{x}$

$10000 + 30000 + \ldots + 100000 = 10(54500)$

$495000 + A = 545000$

$\therefore A = 5000$

$\Sigma y = n\bar{y}$

$175000 + 140000 + \ldots + 91000 = 10(126000)$

$1155000 + B = 1260000$

$B = 105000$

(b) $\Sigma x = 545000 \quad \Sigma y = 1260000 \quad \Sigma xy = 6.1845 \times 10^{10}$

$\Sigma x^2 = 3.7525 \times 10^{10} \quad \Sigma y^2 = 1.65326 \times 10^{11}$

$$B_1 = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma x^2 - \Sigma x \Sigma x} = -0.8724832$$

$$B_0 = \bar{y} - B_1\bar{x} = 173550.3$$

$\therefore g(x) = B_0 + B_1 x$

$= 173550.3 - 0.8724832x$

(c) predicted price $= \hat{y}(75000)$

$= \$105114.1$

(d) $$r = \frac{n\Sigma xy - \Sigma x \Sigma y}{\sqrt{n\Sigma x^2 - \Sigma x \Sigma x}\sqrt{n\Sigma y^2 - \Sigma y \Sigma y}} = -0.9523124$$

$|r| \to 1$ means the relationship between x and y is linear

$r < 0$ means y decreases while x increases

2(a) let group 1 represent the online classroom
let group 2 represent the traditional classroom

$H_0: \mu_1 = \mu_2$
$H_1: \mu_1 \neq \mu_2$

(b) $n_1 = n_2 = 12$

$\bar{x}_1 = 59.83333$ $\bar{x}_2 = 66.08333$

$s_1 = 5.219428$ $s_2 = 5.546096$

pooled variance $s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2} = 31.29924$

$s_p = 5.594593$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = -13.40573$$

at $\alpha = 0.05$ with $df = n_1 + n_2 - 2$

$t_{critical} = qt(1-\alpha/2, n_1+n_2-2)$
$= qt(0.975, 22)$
$= 2.073873$

since $|t| > t_{critical}$, the difference between average maths quiz scores is significant.

(c) While there exists significant difference between the average maths scores, no baseline was established to ensure that the students of both classrooms share the same level of competancy such that the difference between maths score is due to effectiveness of classroom types.

(d) Type 1 error whereby the null hypothesis is incorrectly rejected. In this context, it might be falsely concluded that traditional classroom is more effective than online classroom such that all online classes would be cancelled in future. This would impact adult learners who are unable to attend classes physically due to distance and time constraint.

3(a) H_0: The number of unplanned hospital admissions follows a Poisson Distribution

H_1: The number of unplanned hospital admissions doesn't follow a Poisson Distribution

(b) MLE $\lambda = \frac{\sum xf}{\sum f} = 4.386301$

x	f	$P(X=x)$ = dpois(x, λ)	Expected = $P(X=x) \cdot \sum f$	$\frac{(\text{Expected} - f)^2}{\text{Expected}}$
0	8	0.01245	4.54	2.63053
1	17	0.05459	19.93	0.42997
2	48	0.11973	43.70	0.42245
3	60	0.17506	63.90	0.23785
4	71	0.19197	70.07	0.012357
5	61	0.16844	61.47	0.00358
6	39	0.12312	44.94	0.78440
7	26	0.07715	28.16	0.16542
8	20	0.04230	15.44	1.34754
9	10	0.02061	7.52	0.81453
10	3	0.00904	3.30	0.027734
11	1	0.00360	1.32	0.07590
12	1	0.001318	0.48	0.55983

$$\chi^2 = \sum \frac{(\text{Expected} - f)^2}{\text{Expected}} = 7.51171$$

at $\alpha = 0.05$ and $df = k - p - 1 = 13 - 1 - 1 = 11$

$$\chi^2_{critical} = \text{qchisq}(1-\alpha, df)$$
$$= \text{qchisq}(0.95, 11)$$
$$= 19.67514$$

since $\chi^2 < \chi^2_{critical}$, don't reject H_0

$\therefore$ The number of unplanned hospital admissions follows Poisson Distribution

(c) The number of unplanned hospital admission follows a Poisson Distribution with an average rate of $\lambda = 4.386301$ admissions per day.

3(d) Poisson distribution requires each unplanned admission to be independent. During investigation, care must be taken to ensure all cases are not associated to each other. For example, hospital admissions due to epidemic or a mass accident are not independent of each other. The investigation should exclude data such as this.

3(e) $X \sim Po(\lambda)$

$\Rightarrow E(X) = var(X) = \lambda$

By Central Limit Theorem,

$\bar{X} \sim N(\lambda, \frac{\lambda}{n})$ where $n = 365$

$$
\begin{aligned}
P(\bar{X} > 4.65) &= 1 - P(\bar{X} \leq 4.65) \\
&= 1 - \text{pnorm}(4.65, \lambda, \sqrt{\tfrac{\lambda}{n}}) \\
&= 1 - \text{pnorm}(4.65, 4.386301, 0.1096233) \\
&= 0.00808
\end{aligned}
$$

4(a) i. $E(X) = \sum_{k=1}^{13} k\,P(X=k)$

$= \sum_{k=1}^{13} k\left(\frac{4}{52}\right)$

$= 7$

$var(X) = E(X^2) - [E(X)]^2$

$= \sum_{k=1}^{13} k^2 P(X=k) - 7^2$

$= \sum_{k=1}^{13} k^2\left(\frac{4}{52}\right) - 7^2$

$= 14$

ii. $T_n = X_1 + X_2 + \dots + X_n = \sum_{i=1}^{n} X_i$

$E(T_n) = E\left(\sum_{i=1}^{n} X_i\right)$

$= n\,E(X)$

$= 7n$

$var(T_n) = var\left(\sum_{i=1}^{n} X_i\right)$

$= n.var(X)$

$= 14n$

iii. By Central Limit Theorem,

$E(\bar{X}) = E\left(\frac{T_n}{n}\right) = \frac{1}{n}E(T_n) = 7$

$var(\bar{X}) = var\left(\frac{T_n}{n}\right) = \frac{1}{n^2}var(T_n) = \frac{14}{n}$

$\bar{X} \sim N\left(7, \frac{14}{n}\right)$

$P(X > 7.5) < 0.05$

$1 - P(X \leq 7.5) < 0.05$

$P(X \leq 7.5) > 0.95$

$\frac{7.5-7}{\sqrt{\frac{14}{n}}} > \text{qnorm}(0.95)$

$\sqrt{n} \cdot \frac{7.5-7}{\sqrt{14}} > \text{qnorm}(0.95)$

$\sqrt{n} > \frac{\sqrt{14}}{7.5-7} \cdot \text{qnorm}(0.95)$

$\sqrt{n} > 12.30896$

$n > 151.5104$ $\therefore$ min $n = 152$

4b i. let X be the number of resolved phone calls.

$X \sim Bin(50, 0.6)$

$P(X=30) = \text{dbinom}(30, 50, 0.6) = 0.1145586$

ii. $np = 30$

$n(1-p) = 20$

$\therefore$ normal approximation is valid

let Y be the normal approximation X

$E(Y) = np = 30$

$Var(Y) = np(1-p) = 12$

$P(X=30) = P(29.5 \leq Y \leq 30.5)$

$= \text{pnorm}(30.5, 30, \sqrt{12}) - \text{pnorm}(29.5, 30, \sqrt{12})$

$= 0.1147661$

iii. $P(X<25) = P(X \leq 24)$

$= P(Y \leq 24.5)$

$= \text{pnorm}(24.5, 30, \sqrt{12})$

$= 0.0561756$

4(c) $H_0: W_{after} = W_{before}$

$H_1: W_{after} > W_{before}$

This is a paired test.

$n = 50$ $\bar{X}_{before} = 80$ $\bar{X}_{after} = 85$ $S_{after} = 5$

$d = X_{after} - X_{before}$

$\bar{d} = \bar{X}_{after} - \bar{X}_{before}$

$\Delta d = 0$

$var(d) = var(X_{after} - X_{before})$

$= var(X_{after}) + var(X_{before})$

However, $var(X_{before})$ is not provided.

$\therefore$ unable to proceed with paired testing.